

Nicholas Eisenberg, Ph.D.

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Experience

Senior Data Scientist

July 2022 - Present

Nevada National Security Site

- Trained a Keras model consisting of LSTM and dense layers to forecast future performance of an X-ray machine which resulted in improved maintenance planning and machine repair time
- Developed interactive dashboards using Streamlit and Panel to visually present complex data and provide actionable insights to fellow scientists and stakeholders
- Used Pytorch and Scikit-image to model the spot of an X-ray machine to deconvolve and deblur X-ray images
- Lead investigator for a data engineering project involving the organization and feature extraction from a dataset consisting of 12 years of data from a linear accelerator at the Los Alamos National Lab
- Maintainer of a company-wide internal seismic data-processing Python library hosted on our internal GitLab
- Mentored a student summer-intern and conducted weekly code review meetings to ensure best practices in programming
- Built and managed a MySQL database of seismic sensor metadata
- Utilized Neovim, Tmux, Vim-Slime and SSH to run code on our internal Linux cloud server

Graduate Teaching Assistant

January 2016 - June 2022

University of Nevada, Las Vegas and Auburn University

- Instructed courses in College Algebra and Pre-Calculus
- Led weekly recitation sections for Mathematical Statistics, Real Analysis and Calculus I, II and III
- Managed Auburn University's math tutoring center of 14 tutors

Technical Skills

Programming Languages: Python, SQL, R, Lua, $\text{\LaTeX}$

Libraries: Tensorflow, Keras, Pytorch, Scikit-Learn, Scipy, Plotly, Panel, Pandas, Pyarrow, Sqlalchemy

Developer Tools: Git, Unix/Linux, Neovim, Tmux, SSH

Publications

- 1: Eisenberg, N., Chen, L. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). <https://doi.org/10.1007/s40072-022-00258-6>
- 2: Eisenberg, N., Chen, L. Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ without drift term. Submitted January 2023 to Journal of Theoretical Probability.
Preprint available at <https://arxiv.org/pdf/2209.04771v1.pdf>

Education

Auburn University - Ph.D.

June 2022

Research Area: Probability (Stochastic Processes and Partial Differential Equations)

Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S.

December 2015 / January 2018

Major: Applied Mathematics